

ICT2217 Network Security

Lab Exercise 2.5: TCP SYN Flooding Attack and Defense

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against TCP SYN flooding attack

Required Hardware

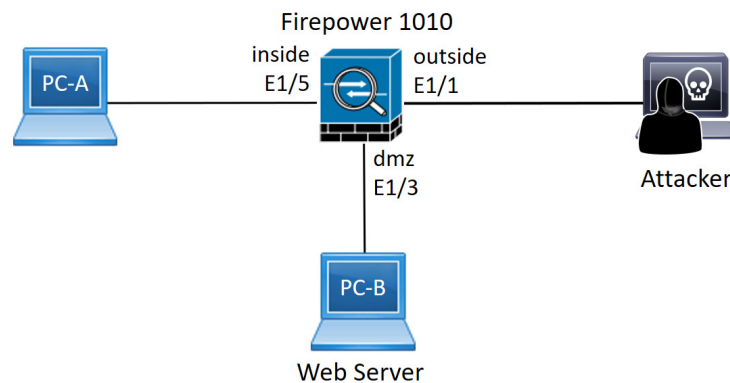
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-RJ45 console cables
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive

PART 1: Conducting TCP SYN flooding attack

1.1 Setup the network topology as shown:

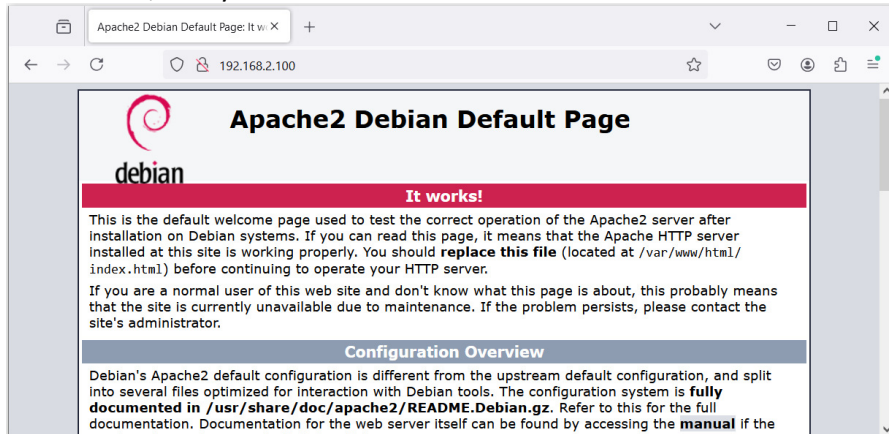


- 1.2 Configure the interfaces of the Firepower/ASA firewall with suitable IP addresses, subnet masks, names and security-levels.
- 1.3 Boot up Attacker PC and Web Server PC in Kali Linux, PC-A may be in Windows.
- 1.4 Configure suitable IP addresses, subnet masks and default gateways to the PCs.
- 1.5 Apply suitable ACL on the firewall to allow Attacker to visit PC-B Web Server hosted at DMZ, but not PC-A inside enterprise LAN.

- 1.6 At PC-B Web Server, start Apache web server and disable SYN cookie as shown:

```
(kali@kali)$ sudo systemctl start apache2
(kali@kali)$ sudo sysctl -w net.ipv4.tcp_syncookies=0
```

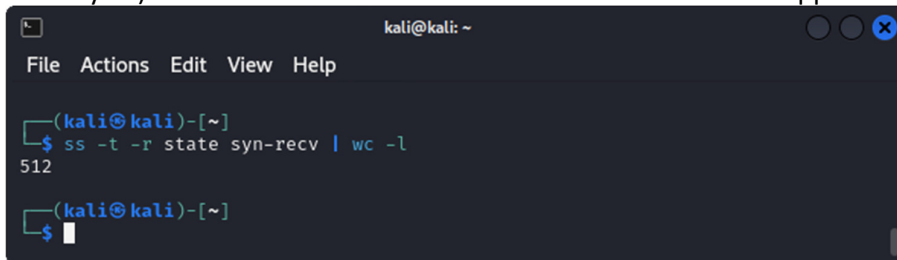
- 1.7 When done, verify that PC-A and Attacker are able to browse the Web Server:



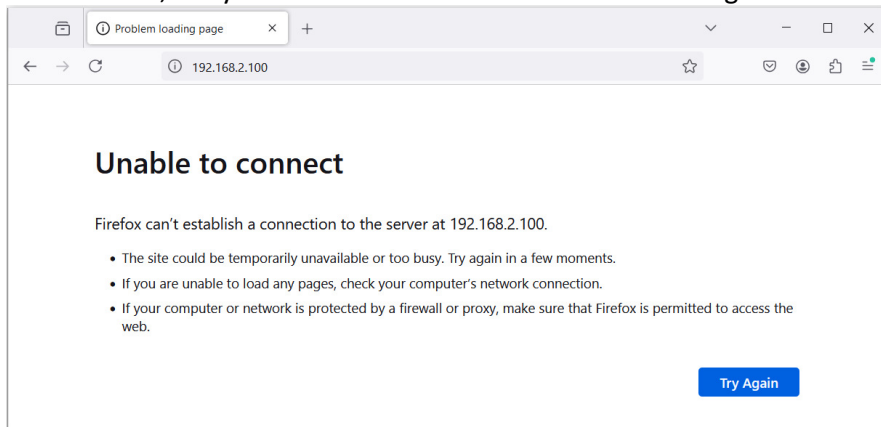
- 1.8 Next, conduct TCP SYN flooding attack from the Attacker's PC to the Web Server by running the provided python scapy script as follows:

```
(kali@kali)$ sudo python tcpsynflood target-ip target-port
```

- 1.9 At the Web Server, show the count of TCP connections in SYN-RECV state (aka embryonic) until it reaches 512 which is the default maximum supported as shown:



- 1.10 Now at PC-A, are you able to browse the Web Server during the attack?



PART 2: Defending against TCP SYN flooding attack using MPF service policy

- 2.1 Configure embryonic connection limits using MPF as discussed.
- 2.2 When done, conduct TCP SYN flooding attack from the Attacker's PC to the Web Server again.
- 2.3 At the Web Server, is the count of TCP connections in SYN-RECV state (aka embryonic) able to reach 512 as in Step 1.9?

```
(kali@kali)$ ss -t -r state syn-recv | wc -l
```

- 2.4 Now, is PC-A able to browse the Web Server during the attack?
- 2.5 Do you think the ASA MPF service policy that you have configured is useful against TCP SYN flooding attack?